



Internship Report On Data Science with Python

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Student's Declaration

I **Manish Kumar Mahato**, a student of **BCA (AI & ML)** program, Roll No. **024102020027** of the Department of **Computer Application** College do hereby declare that I have completed the mandatory internship in EISystems Technologies under the faculty guideship of **Dr.Rohit Kumar**, Department of **Computer Application**, **Haridwar University**.

(Signature and Date)

Endorsements

SIGNATURE

Bhanu Pratap Singh
Department of Computer Application
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Nomenclature / Notations

Serial No	Notations	Description
1	Df	DataFrame (pandas)
2	x	Features / Independent Variables
3	y	Target Variable / Dependent Variable
4	coef_	Model Coefficients
5	intercept_	Model Intercept
6	SVC	Support Vector Classifier

Executive Summary

This report details the successful completion of a short-term internship at EISystems Services, where the primary focus was on developing a machine learning model for house price prediction. The objective was to build a robust predictive model using Linear Regression, leveraging Python and key libraries like pandas and scikit-learn. The training provided a strong foundation in data science principles and model implementation. The project involved data loading, preprocessing, model training, and evaluation. The outcomes achieved include the development of a functional and interpretable model, a deeper understanding of regression analysis, and practical experience in the machine learning workflow.

Overview of Organization

Introduction of the Organization

EISystems Services (EISys) is a leading Indian technology training and education provider. It offers workshops, internships, and project-based learning in fields like Artificial Intelligence, Machine Learning, Internet of Things (IoT), Robotics, Cybersecurity, and Automobiles. The organization has trained over 50,000 students directly and collaborated with IITs, NITs, and more than 100 top engineering colleges across India.

Vision, Mission and Values

- **Vision:** To empower students with practical, industry-ready technical skills.
- **Mission:** To bridge the gap between academic knowledge and real-world application through hands-on training and mentorship.
- **Values:** Innovation, excellence, knowledge-sharing, and ethical learning.

Policy in Relation to the Intern Role

EISystems emphasizes experiential learning by involving interns in live projects that reflect real-world challenges. The organization encourages creativity, problem-solving, teamwork, and professional discipline, ensuring interns gain both technical expertise and industry readiness.

Project Summary

1. Idea Behind the Project

The primary idea behind this project was to apply the principles of linear regression, a fundamental machine learning algorithm, to a real-world problem: predicting house prices. By using a dataset containing various features of a house, such as its area, number of bedrooms, and age, the goal was to create a model that could accurately estimate its market value.

2. About the Project

This project involves building a supervised machine learning model to predict a continuous value (price) based on multiple input features. We used the Linear Regression algorithm, which is well-suited for this type of task. The project follows a standard machine learning workflow: data loading, feature selection, model training, and prediction.

3. Software Used

- **Python:** The primary programming language used.
- **Jupyter Notebook:** The interactive environment for developing and running the code.
- **Pandas:** A library for data manipulation and analysis.
- **Scikit-learn:** A robust library for machine learning in Python, used for the Linear Regression model.

4. Technical Apparatus Requirements

- A computer with Python installed (version 3.x or higher).
- Required Python libraries: pandas and scikit-learn.
- A CSV file containing the dataset (e.g., "Multi.csv").

5. Results and Working

The linear regression model was successfully trained on the provided dataset. The model's coefficients and intercept were calculated, providing insights into the relationship between each feature and the house price. The model was then used to make a prediction for a new, unseen data point, demonstrating its predictive capability.

6. Research Done

The research involved understanding the core concepts of linear regression, including its mathematical principles, assumptions, and how to interpret the model's coefficients. We also researched the proper use of the pandas and scikit-learn libraries to handle data and build the model effectively.

Data Flow Diagram / Process Flow

The process flow of the software project is a linear sequence of steps:

1. **Data Ingestion:** The "Multi.csv" dataset is read into a pandas DataFrame.
2. **Feature and Target Separation:** The DataFrame is split into two parts: a feature matrix (x) containing area, bedrooms, and age, and a target vector (y) containing price.
3. **Model Instantiation:** An instance of the LinearRegression model is created from the scikit-learn library.
4. **Model Training:** The `model.fit()` method is called to train the model using the feature data (x) and target data (y).
5. **Prediction:** The trained model is used to predict the price for a new data point by calling `model.predict()`.
6. **Output:** The model's coefficients, intercept, and the final prediction are printed to the console.

Code / Program with Supported Screenshots

The following is the complete Python code used for the project, with comments explaining each step.

```
# Import the pandas library for data manipulation
import pandas as pd

# Read the data from a CSV file named "Multi.csv" into a pandas DataFrame
df = pd.read_csv("Multi.csv")

# Display the DataFrame to see the data
print("The full DataFrame:")
print(df)

# Define the features (independent variables) for the model
# The features are 'area', 'bedrooms', and 'age'
x = df[["area", "bedrooms", "age"]]

# Display the features DataFrame
print("\nFeatures (x):")
print(x)

# Define the target variable (dependent variable)
# The target is 'price'
y = df.price

# Display the target series
print("\nTarget (y):")
print(y)

# Import the LinearRegression model from scikit-learn
from sklearn.linear_model import LinearRegression

# Create an instance of the LinearRegression model
model = LinearRegression()

# Train the model using the features (x) and target (y)
model.fit(x, y)

# Print the coefficients of the model
# These represent the weights for each feature
print("\nModel coefficients:")
print(model.coef_)

# Print the y-intercept of the model
print("\nModel intercept:")
print(model.intercept_)

# Make a prediction for a new data point with area=5000, bedrooms=3, and age=5
# To avoid the UserWarning, we create a new DataFrame for the prediction
new_data = pd.DataFrame([[5000, 3, 5]], columns=["area", "bedrooms", "age"])
print("\nPrediction for a new data point:")
print(model.predict(new_data))
```

Input / Output with Datasets & Supported Screenshots

The input data for this project is a CSV file named Multi.csv. The output is the predicted price for a new data point.

	A	B	C	D
1	area	bedrooms	age	price
2	2600	3	20	550000
3	3000	4	15	565000
4	3200	3	18	610000
5	3600	3	30	595000
6	4000	5	8	760000

```

✓ TERMINAL
PS D:\New folder> python -u "d:\New folder\House_Price_Pred
iction_using_Linear_Regression.py"
The full DataFrame:
   area  bedrooms  age  price
0  2600         3   20  550000
1  3000         4   15  565000
2  3200         3   18  610000
3  3600         3   30  595000
4  4000         5    8  760000

Features (x):
   area  bedrooms  age
0  2600         3   20
1  3000         4   15
2  3200         3   18
3  3600         3   30
4  4000         5    8

Target (y):
0   550000
1   565000
2   610000
3   595000
4   760000
Name: price, dtype: int64

Model coefficients:
[ 137.25 -26025.  -6825. ]

Model intercept:
383724.99999999994

Prediction for a new data point:
[957775.]
PS D:\New folder>

```

Images / Video Links

High Quality Images and Videos Google Drive Link :-

https://drive.google.com/drive/folders/1daSQ0QzOVHJxnp5n2M2Azp7pkvN679-1?usp=drive_link

References

1. **Scikit-learn Documentation:** <https://scikit-learn.org/stable/>
2. **Pandas Documentation:** <https://pandas.pydata.org/docs/>

Student Self Evaluation of the Short-Term Internship

1) Oral communication	3
2) Written communication	5
3) Initiative	4
4) Interaction with staff	3
5) Attitude	3
6) Dependability	4
7) Ability to learn	5
8) Planning and organization	5
9) Professionalism	4
10) Creativity	4
11) Quality of work	4
12) Productivity	4
13) Progress of learning	4
14) Adaptability to organization's culture/policies	5
15) OVERALL PERFORMANCE	4

Rating Scale: 5 will be Best while 1 will be Worst

Signature of the Student

Annexure 1

Daily Activity Report

Week No: 1

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 1	Orientation to the company and the internship project. Started with learning Python basics, including syntax, data types, and fundamental functions.	Gained a strong foundation in Python programming, which is essential for the project.	
Day 2	Setup of the development environment. Installation of Python, Jupyter Notebook, pandas, and scikit-learn. Initial research on Linear Regression.	Successfully set up all necessary tools and began foundational research.	
Day 3	Sourcing and understanding the dataset. Reading the "Multi.csv" file into a pandas DataFrame. Initial data exploration and analysis.	Learned how to handle data with pandas and gained insights into the dataset's structure.	
Day 4	Data cleaning and preprocessing. Separating the dataset into features (x) and the target variable (y)	Developed skills in data preparation, a critical step in the machine learning pipeline.	
Day 5	Building the Linear Regression model. Importing the LinearRegression class and training the model using the .fit() method.	Successfully implemented and trained a basic machine learning model.	

Week No: 2

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 6	Discussion on model evaluation metrics. Calculation and interpretation of R2 score and Mean Squared Error (MSE).	Gained an understanding of how to assess the performance and accuracy of a machine learning model.	
Day 7	Data visualization. Creating charts to understand feature distributions and their correlation with the target variable.	Learned to use data visualization to uncover patterns and relationships within the data.	
Day 8	Research and application of different regression models. Explored Lasso and Ridge regression for comparison.	Expanded knowledge beyond Linear Regression to more advanced techniques.	
Day 9	Initial draft of the project report sections, including the Executive Summary and Project Summary. Initial draft of the project report sections, including the Executive Summary and Project Summary.	Developed skills in technical writing and project documentation.	
Day 10	Review of Week 2 progress with the mentor. Discussion of model performance and potential improvements.	Received valuable feedback and guidance, ensuring the project was on track.	

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Annexure 1

Daily Activity Report

Week No: 3

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 11	Refinement of the code for clarity and commenting. Adding more detailed explanations within the script.	Improved coding practices by focusing on readability and maintainability.	
Day 12	Working on the Data Flow Diagram and Process Flow sections of the report.	Created visual representations of the project workflow.	
Day 13	Gathering and organizing screenshots of the code, input dataset, and output.	Learned the importance of clear and organized visual documentation.	
Day 14	Drafting the Introduction, References, and Student Self-Evaluation sections of the report.	Completed a significant portion of the report's structure.	
Day 15	Finalizing the report draft for review. Proofreading for grammar and consistency.	Practiced attention to detail in technical report writing.	

Week No: 4

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 16	Incorporating feedback from the mentor on the report draft. Making necessary revisions to the content and formatting.	Experienced a professional feedback and revision cycle.	
Day 17	Preparing for the final project presentation. Creating a presentation deck to visually explain the project's key aspects.	Developed public speaking and presentation skills.	
Day 18	Rehearsing the final presentation and refining the talking points.	Polished presentation delivery for clarity and impact.	
Day 19	Final project submission day. Submitting the code, report, and presentation.	Gained experience in the formal submission process for a project.	
Day 20	Debriefing session with the mentor. Final feedback and discussion on overall internship experience.	Reflected on the internship experience and received career guidance.	

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Annexure 2

Weekly Progress Report

Week No: 4

Week(s)	Summary of Weekly Activity
Week 1	During this week, I successfully set up my development environment, conducted foundational research on linear regression, and successfully trained the model on the dataset.
Week 2	This week, I focused on evaluating the model's performance using key metrics, explored data visualization techniques, and began drafting the project report.
Week 3	I focused on documenting the project this week. I refined the code, created a data flow diagram, gathered all necessary screenshots, and completed the main body of the project report.
Week 4	This week was dedicated to finalizing the project. I incorporated feedback on my report, created and rehearsed a presentation, and prepared for the final submission and debriefing with my mentor.